

Ethan Duong

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EDUCATION

University of California, Berkeley

Berkeley, CA

Bachelor of Arts in Computer Science & Data Science

Expected Graduation: May 2029

Relevant Coursework: Structure and Interpretation of Computer Programs, Data Structures, Principles and Techniques of Data Science, Discrete Mathematics and Probability, Linear Algebra, Calculus I/II/III

TECHNICAL SKILLS

Languages: Python, Java, C, C++, Scheme, SQL, HTML, CSS

Concepts: Object-Oriented Programming, Data Structures, Algorithms, Functional Programming

Tools: Git, GitHub, Jupyter Notebook, Visual Studio Code, IntelliJ IDEA

EXPERIENCE

Assistant Manager

November 2023 – August 2025

Kumon

San Jose, CA

- Tutored **50+** students in Mathematics and English, adapting instruction to different skill levels.
- Tracked student performance data and adjusted individualized learning plans to improve outcomes.
- Managed daily center operations, including worksheet grading, scheduling, and parent communication.
- Streamlined workflow for grading and organization to maintain efficiency during high-volume sessions.

PROJECTS

BYOW: Build Your Own World Game Engine | *Java, OOP, Algorithms*

April 2026

- Implemented a procedurally generated 2D world using seeded randomness for deterministic generation.
- Built a tile-based engine with room generation, hallway connectivity, and path connectivity constraints.
- Developed real-time avatar movement with collision detection and keyboard and mouse input handling.
- Designed save and load functionality by serializing full game state for exact reproducibility.

Scheme Interpreter | *Python*

November 2025

- Implemented an interpreter supporting recursion, lambda functions, and lexical scoping.
- Built a tokenizer and parser to process user input and evaluate nested Scheme expressions correctly.
- Designed an interactive REPL to read, evaluate, and print user input with continuous execution.

CATS (Typing System) | *Python*

October 2025

- Built a typing tool that measures speed and accuracy by comparing user input against reference text.
- Added an autocorrect feature using word similarity to catch and fix typing mistakes as users type.
- Organized modular code to support different typing modes and make future features easier to add.

PiE Robotics Competition | *VEX, Hardware Design*

March 2024 – April 2024

- Built and tested a VEX robot for an obstacle-based competition using iterative refinement.
- Improved drivetrain and mechanism reliability through repeated testing and small design adjustments.
- Documented design changes and test results to help guide team decisions during development.

EXTRACURRICULAR ACTIVITIES

Pioneers in Engineering

August 2025 – Present

University of California, Berkeley

Berkeley, CA

- Created features for a real-time scoreboard for high school robotics competitions across the Bay Area.
- Integrated match data with the scoreboard to ensure accurate and timely updates during competitions.
- Collaborated with software, hardware, and outreach teams to support competition operations.